

Honours Thesis - Simulation of Squeezed Light Generation in an Optical Resonator

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CONTENTS

I. PROJECT MOTIVATION

MacRae's Outline:

Quantum fluctuations of the electromagnetic field impose a fundamental limit on measurement sensitivity, known as the standard quantum limit (SQL). Exotic quantum optical states known as "squeezed light" can surpass the SQL. Squeezed states are a hot topic in research, particularly in quantum communication, metrology, and optical quantum computing. This project will explore theoretical models predicting strong squeezing when atoms are placed inside an optical cavity. The student will develop and run numerical simulations to investigate the conditions for optimal squeezing, using tools from quantum optics and cavity QED theory. Through this work, the student will gain experience in programming, numerical methods, and the theoretical framework of quantum light-matter interactions.

The past hundred years since the formulation of quantum mechanics have led to unrivaled technological progress, especially in recent years. With greater understanding of and ability to control quantum phenomena further possibilities are becoming available. These include quantum key distribution (QKD), quantum communication, and quantum computation, to name a few.

Any emerging technology tends to follow the rule ascribed to increasing system complexity sometimes stated as 'more is different' with reference to the 1972 paper by Physicist Philip W. Anderson [?]. This especially applies to that of quantum networks. As greater quantum control is achieved, greater quantum networking capabilities will be required, thereby necessitating the development of improved methods for transmitting quantum information.

Currently, a hurdle in developing quantum networks is transmitting quantum information with a sufficiently long coherence time such that the information produced may actually be used. This is especially true if the information is to be transmitted across vast distances through optical fibres, such as China has done in a 4600 kilometer land-based network [?].

One dimension of this problem is that even if a coherent optical field is produced, it may not be at a frequency for which optical fibres have a low-loss window. Thus, effective frequency conversion methods which preserve quantum information must be developed.

This thesis includes a theoretically viable method for frequency conversion of two entangled electromagnetic

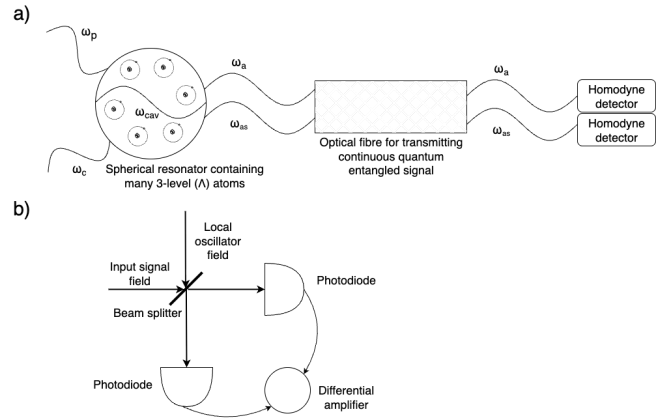


FIG. 1: Suggested setup of a real-world implementation for the four wave mixing process described with (a) two continuous beams entangled beams (continuous-variable entanglement) at frequencies ω_p and ω_c which undergo a four wave mixing process within a spherical resonator tuned to frequency ω_{cav} filled with cooled three-level atoms to produce Stokes and Anti-Stokes beams at frequencies ω_a and ω_{as} , respectively. The resulting beams are then may then be sent through an optical fibre to homodyne detectors. (b) Homodyne detector layout.

fields (continuous-variable entanglement). The key idea is to send the entangled beams into an optical spherical resonator filled with cooled three-level atoms (Λ -system) such that a four wave mixing (FWM) process occurs to produce two entangled output fields. If the output frequencies are tuned such that they incur low-loss within an optical fibre, then the signals could theoretically be transmitted over long distances with existing technology. For a diagrammatic representation of this process, see figure ??.

TODO - Should squeezed states be mentioned here, or is it fine to leave them for later?

The outline of this project can be described in three stages: formulating up the system Hamiltonian, computationally solving for energy eigenstates of the Hamiltonian, and analyzing the results to find how changing specific parameters may change the system behavior.

II. BACKGROUND INFORMATION

A. Optical Non-Linearities

B. Electromagnetically Induced Transparency

In order to gain an intuition on electromagnetically induced transparency (EIT), we will start with a classical toy model of the atom. As is a common theme in physics, a hydrogenic atom in a time dependent electromagnetic field can be approximately modelled as a driven harmonic oscillator. So rather than dealing with quantized photons, we will consider a classical oscillating electromagnetic fields with frequencies nearly resonant with those of the atomic transitions.

To do this, a few assumptions will be made [?]: first, the wavelengths of the light sources interacting with the atom have wavelengths much greater than the Bohr radius of the atom ($\lambda \gg a_0$); second, the detunings of incoming beams are small in proportion to the transition frequencies considered. These two assumption happen to be consistent with one another.

The first assumption allows for us to approximate the electromagnetic field experienced by the atom as constant in space but varying in time since the change in electromagnetic field strength between the two ends the atom in the transverse wave direction is relatively small.

There exists the relation between the electric and magnetic field strengths of light which arises from Maxwell's equation [?]

$$B = \frac{E}{c}. \quad (1)$$

The Lorentz force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

where q is charge, and $\mathbf{v}$ is particle velocity. Using the Bohr model for approximation, we have that quantized electron velocities for a hydrogenic atom have magnitude

$$v_n = \alpha \frac{Z}{n} c \quad (3)$$

where α is the fine structure constant, Z is the atomic number, and n is the principle quantum number [?]. We can therefore see by rough approximation that we have

$$|\mathbf{v} \times \mathbf{B}| \lesssim \alpha Z c \frac{E}{c} = \alpha Z E. \quad (4)$$

We are considering Rubidium with atomic number $Z = 37$, but this neglects the effects of screening caused by the electrons in inner shells of the atom. Electron position is distributed in a cloud about the atom for any orbital, though we can expect the 36 electrons of inner shells between the valence electron (singular in its shell) and the nuclear charge of $37e$ to roughly cancel $36e$ of charge

such that the valence electron experiences about $1e$. By this line of reasoning we have

$$|\mathbf{v} \times \mathbf{B}| \lesssim \alpha E. \quad (5)$$

Thus, the force due to the external magnetic field is about $1/137$ that of the external electric field. So we may neglect the external magnetic field for simplification.

$$\mathbf{F} \approx q\mathbf{E} \quad (6)$$

All of these approximations can be justified from the perspective that this model is a first estimate for system behavior which will be refined and iterated upon.

This brings us to a physicist's favorite abstraction: the harmonic oscillator. The potential of which the valence electron experiences can be expanded in a Taylor series and approximated as quadratic [?]. The mass of the nucleus is much greater than the electron mass, so we may approximate the nucleus to be fixed. So we have a mass with some equilibrium distance from a fixed point under a quadratic potential, i.e. a simple harmonic oscillator. We may then add in our external time-dependent driving force to obtain a driven harmonic oscillator.

When a charge oscillates, it emits photons as its movement produces a changing electric and magnetic field. So when our system is driven on resonance, then an electromagnetic field of that same frequency will be produced. Here lies the idea behind electromagnetically induced transparency. By driving a cloud of these hydrogenic atoms on resonance, the field essentially 'passes through' the medium.

C. Structure of atomic spectra

Atomic states for the electron of a hydrogenic atom are enumerated by the quantum numbers n (principle quantum number), l (angular momentum quantum number), m_l (magnetic quantum number), and m_s (spin quantum number) [?]. **TODO - Talk about hydrogenic energy levels for unperturbed idealized hydrogen, then fine and hyperfine structure such that it leads nicely into 3-level atoms.**

D. Atomic transitions

In a hydrogenic atom, the transitions between states are mediated by excitations in the electromagnetic field (i.e. photons). It is crucial in atomic physics to understand the probabilities of these atomic state transitions.

The interaction between an atom and the electromagnetic field is not a linear interaction, creating difficulty in finding an exact solution. To obtain approximate behavior, first order perturbation theory can be applied to dominating effects of this interaction [?]. From this, one obtains the transition dipole matrix with elements

$$\mu_{j,i} = \langle j | \hat{\mathbf{d}} | i \rangle = e \langle j | \hat{\mathbf{r}} | i \rangle \quad (7)$$

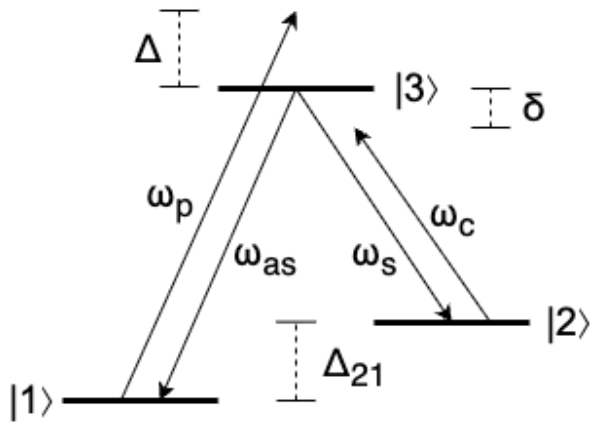


FIG. 2: TODO

where $\hat{\mathbf{d}} = e\hat{\mathbf{r}}$ is the dipole transition operator and $\hat{\mathbf{r}}$ is the position operator [?].

If $\mu_{j,i} = 0$, then the transition between states $|i\rangle$ and $|j\rangle$ is a ‘forbidden’ dipole transition, otherwise the transition is an allowed dipole transition. This criterion enforces the dipole selection rules that

$$\Delta l = \pm 1, \Delta m_l = 0, \pm 1. \quad (8)$$

It is important to note that these transition rules are not always perfectly enforced. Since perturbation theory was applied to obtain them, there exist transitions which disobey these rules, however, these transitions occur with significantly decreased probability.

E. Three-Level Atom

F. Four Wave Mixing

G. Squeezed States

According to Heisenberg’s uncertainty principle, the product of the standard deviations of two canonically conjugate observables, x and p , must satisfy

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (9)$$

For light of sufficient intensity, one can consider its amplitude and phase as canonically conjugate variables [?]. For a generically produced beam, one would expect the standard deviations of amplitude and phase to be roughly equal. If measurement were to be performed in a single quadrature (amplitude for example), then reducing the uncertainty in that quadrature would result in a more precise overall reading.

This leads to the idea of a squeezed state in which one quadrature has its uncertainty decreased below the shot noise level at the expense of the other, such that the

uncertainty principle is still observed. It has been shown that optical squeezed states can be produced through a FWM process in optical cavities [?].

H. Continuous Variable Quantum Information

If multiple beams are entangled such that a measurement of one of these variables on one beam yields information about the other beams, then we have what is called continuous variable entanglement [?]. In this way, quantum information can be stored, processed, and transmitted using continuous beams rather than with individual photons.

I. How Sub-Poissonian Light and Photon Anti-Bunching Effects Affect Continuous Variable Measurements

Continuous variable measurements allow for small quantum fluctuations to be less significant in measurement procedures when compared to individual photon entanglement, though there still exist challenges in performing measurements. Continuous variable measurements are typically performed with homodyne detectors [?], a type of interferometer which compares some input beam against a controlled probe beam and takes the difference between their readings [?].

Light is quantized and beam intensity is not that high. Individual photons arrive at the detector in a Poisson process, resulting in shot noise for the detector [?]. Due to this, the reading of the input beam can be swayed by individual photon arrival events. It would be highly desirable for the input beam to follow sub-Poissonian statistics (with the same mean as a Poisson distribution but a lower variance) [?] and for the beam to display photon anti-bunching (arrival events are distributed more evenly as opposed to having intensity spikes) [?].

It has been shown that applying cavity EIT to three-level atoms can produce sub-Poissonian light [?], and that applying FWM to three-level systems can produce photon anti-bunching effects [?]. Combining these techniques theoretically allows for the information of continuous variable beams to be more reliably transmitted to some measurement device, such that readings fall below the shot-noise level.

J. Fock Basis

K. Master Equation

III. THE FIRST STEPS

In order to work up to modelling a three-level atom in the Fock basis for photons with dissipative Lindbladian operators, a simpler model will be taken as a starting

point. We will treat our atom as a three-level system with discrete states while lights behaves as a continuous field such that we do not need to discretize light in the Fock basis of photon number.

The polarization vector of the atomic cloud may be given by

$$\mathbf{P} = N \langle \hat{\mathbf{d}} \rangle$$

where $\langle \hat{\mathbf{d}} \rangle = \epsilon_0 \chi \mathbf{E}$ (10) where N is atomic density, $\hat{\mathbf{d}}$ is the averaged dipole moment, and χ is the magnetic susceptibility [?]. One can find that for a three-level atom (TODO derive or cite?)

$$\chi = \frac{-Nd^2}{\epsilon_0 \hbar} \frac{\delta}{|\Omega_c|^2 - \delta(\Delta + i\Gamma)} \quad (11)$$

where TODO - Define detunings and dissipation rate along with a figure for a three-level atom.

The Hamiltonian of the three-level atom can be given as

$$H_{atom} = \text{TODO}(\text{diagonal} - \text{slides}) \quad (12)$$

and the Hamiltonian of the light-matter interaction can be given as

$$H_{atom} = \text{TODO}(\text{see slides}) \quad (13)$$

The dissipative operators...

Solve master equation with...

Here's the plot...

A. Meeting 2 with MacRae

1. Hamiltonian Derivation with light as a classical field

In order to work up to modelling a three-level atom in the Fock basis for photons with dissipative Lindbladian operators, a simpler model will be taken as a starting point. We will treat our atom as a three-level system with discrete states while lights behaves as a continuous field such that we do not need to discretize light in the Fock basis of photon number.

To construct the Hamiltonian, it is advantageous to first express it as a sum of the contributions to be considered

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction} + \hat{H}_{Light}. \quad (14)$$

Since we are considering the simplified system where light acts as a continuous field, all we care about is how that field interacts with the atom. So at this point, we choose to neglect the term $\hat{H}_{Light}$ to obtain

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction}. \quad (15)$$

Let $\hat{\sigma}_{i,j} = |i\rangle\langle j|$ express the atomic transition operators which operate on state vectors to transition from state j to i . For example,

$$\hat{\sigma}_{1,3} = |1\rangle\langle 3| = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (16)$$

and acting this operator on basis state $|i\rangle$ yields

$$\hat{\sigma}_{1,3}|i\rangle = |1\rangle\langle 3|i\rangle = \delta_{3,i}|1\rangle \quad (17)$$

where δ denotes the usual Kronecker delta.

TODO

3-level system frequency 2-3 = Ω_c . $\Gamma_{3 \rightarrow 2}$ and $\Gamma_{3 \rightarrow 1}$ emission. $\hat{\sigma}_{i,j} = |i\rangle\langle j|$. $\Delta\omega_s - \omega_{3,1}$. $\delta = \Delta_c - \Delta$

$$\hat{H} = \hat{H}_a + \hat{H}_{LightAtom} + \hat{H}_{Light}$$

Neglect $\hat{H}_{Light}$

$$\hat{H}_a = \hbar(\omega_{3,1} - \omega_{3,1})\hat{\sigma}_{2,2} + \hbar\omega_{3,1}\hat{\sigma}_{3,3}$$

$$\hat{H}_{LightAtom} = -\hat{\mathbf{d}}_{1,3} \cdot \mathbf{E}_s - \hat{\mathbf{d}}_{2,3} \cdot \mathbf{E}_c$$

Expand dipole operators and E in terms of complex exponentials.

$$\hbar\Omega_s = \mathbf{d}_{1,3} \cdot \mathbf{E}_s$$

$$\hbar\Omega_c = \mathbf{d}_{2,3} \cdot \mathbf{E}_c$$

Plug in Ω_i values into hamiltonian, replacing dipoles and E

We need the interacting picture

$$\hat{H} = \hat{H}_0 + \hat{H}_I$$

$$\hat{H}_0 = \omega_s\hat{\sigma}_{3,3} + (\omega_s - \omega_c)\hat{\sigma}_{2,2}$$

$$\frac{\hat{H}_I}{\hbar} = \frac{\hat{H} - \hat{H}_0}{\hbar} = (\omega_{3,1} - \omega_s - \omega_s + \omega_{3,2})\hat{\sigma}_{2,2} + (\omega_{3,1} - \omega_s)\hat{\sigma}_{3,3} + \hat{H}_{LightAtom}$$

Plug in Δ and δ then $\hat{H}_{LightAtom}$ to obtain interaction hamiltonian

$$\hat{H}_I = \begin{pmatrix} 0 & 0 & \Omega_s e^{i\omega_s t} \tilde{\Omega}_s e^{i\omega_s t} \\ 0 & \delta & \Omega_c e^{i\omega_c t} \tilde{\Omega}_c e^{i\omega_c t} \\ c.c & c.c & \Delta \end{pmatrix}$$

Then

$$V = e^{-i\hat{H}_0 t} \hat{H}_I e^{i\hat{H}_0 t}$$

Notice that $e^{-i\hat{H}_0 t}$ is diagonal so this is simple

Multiply through, then apply rotating wave approximation to obtain

$$\hat{V} = \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & \delta & \Omega_c \\ c.c & c.c & \Delta \end{pmatrix}$$

2. a

Density matrix times an operator

$$\hat{\rho}\hat{A} = |\psi\rangle\langle\psi|\hat{A}$$

$$\hat{B} = I\hat{B} = \sum_c |c\rangle\langle c| \hat{B} |c\rangle\langle c|$$

Do some math, then we find that

$$\langle \hat{A} \rangle = \text{Tr}[\hat{\rho}\hat{A}]$$

For linear media, the polarization is $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0(1 + \xi)\mathbf{E} = \epsilon \mathbf{E}$ and $\mathbf{P} = N \langle \hat{\mathbf{d}} \rangle$

$$1 + \xi = \frac{\epsilon}{\epsilon_0}$$

For non-magnetic materials $\mu \approx \mu_0$, so

$$1 + \xi = \frac{\epsilon}{\epsilon_0} = \frac{\epsilon\mu}{\epsilon_0\mu_0} = \left(\frac{c}{v}\right)^2 = n^2$$

Something like this, check all these derivations

Write up derivation to the point that we obtain

$$\xi = \frac{N\langle d \rangle}{\epsilon_0 \hbar \Omega}$$

$$\langle \mathbf{d} \rangle = Tr[\hat{\rho}(\hat{\mathbf{d}}\hat{\sigma}_{i,j} + h.c.)]$$

Expand out (Find a textbook with this) and we obtain

$$\xi = \frac{Nd^2}{\epsilon_0 \hbar \Omega} \rho_{i,j}$$

The density matrix should be a unitary

This

IV. HAMILTONIAN FORMULATION

V. COMPUTATIONAL METHODS

VI. RESULTS

VII. DISCUSSION

A. Conditions for Optical Squeezing